Chapter 2 Review

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This is a document to review a couple concepts of FOL.

Why FOL-tableau would not work if we used a domain other than the domain of the denumerable parameters

This fact can be exhibited by looking back at Problem 2 of Chapter 9 of the Beginners Guide [Smullyan, 2014]. The question is to prove that if I satisfies a set S, then for any $\delta \in S$ and any parameter a that is new to S, the set $S \bigcup \delta(a)$ is satisfied by some extension. Because a is a parameter, this is really easy: the way parameters were defined allowed us to assign to them *any* element of the domain under any interpretation whatsoever; this is why this works for the parameters. However, if we used some other domain, given that I satisfies S, we don't even know that the elements of that domain are in the domain under I. And even if we did, we still don't know if $\delta(a)$ would be true under some extension of I, as a is no longer a parameter.

Clarifying the role of predicates in FOL-valuations

Recall that we have the following definition included in the definition of a FOL-valuation (in the domain of the parameters):
B_1: The sentence $\sim X$ receives the value t under v iff $v(X) = f$; and accordingly $v(\sim X) = f$ iff $v(X) = t$.
where X is a sentence. This means that we can still assign t or f to predicates, as they are sentences (as long as its set of n-tuples $(a_1, ..., a_n)$ does not include any variables). However, we no longer have the notion of an interpretation $I(P)$

over a predicate P. But this is fine, as we can show that given any interpreation I, there is one and only one valuation v such that, for every sentence X, X is true under I iff X is true under v. This is just the valuation that assigns t to those and only those sentences that are true under I. Also, we can show that, for any valuation v, there is one and only one interpreation I such that, for every sentence X, X is true under I iff X is true under v. This is just the interpretation (in the domain of the parameters) such that, for any predicate P of degree n, take $I(P)$ to be the relation that $P_{a_1,\ldots,a_n} = t$ (in the denumerable domain of the parameters) under v.

Exercise from Chapter 8 of the Beginners Guide

Base Case: On the one hand, if F (in this case, an n-ary predicate) is true under I_2, this means, by definition of $I_2(P)$, F' is true under I_1. On the other hand, if F' is true under I_1, then this would make F be true under I_2, by the definition of $I_2(P)$.
Induction: Suppose that, for all numbers less than n, the result holds for every formula of these said numbers. Thus, if F is of type α, α_1, α_2 are true iff α_1', α_2' are true, respectively, meaning F is true iff F' is true. If F is of type β, β_1 is true iff β_1' is true, and likewise for β_2 and β_2'. Thus, F is true iff F' is true. If F is of the form $\forall x\phi(x)$, $\phi(a)$ is true iff $\phi(a)'$ is true, for every parameter a. Thus, $\phi(a)$ is true for every parameter a iff $\phi(a)'$ is true for every parameter a, thus meaning F is true iff F' is true. Lastly, if F is of the form $\exists x\phi(x)$, $\phi(a)$ is true iff $\phi(a)'$ is true, for every parameter a. Thus, $\phi(a)$ is true for at least one parameter a iff $\phi(a)'$ is true for at least one parameter a, thus meaning F is true iff F' is true.